

IRONEYES (FORGE / IRONLOOP) Integrated Master Project Report

SIH 2026 | Problem Statement: SIH26213 | Category: Hardware (Fitness & Sports)

System Title: FORGE / IRONLOOP — Closed-Loop Velocity-Based Training, Load Management, Movement Feedback, and Coach-Assisted Recovery System

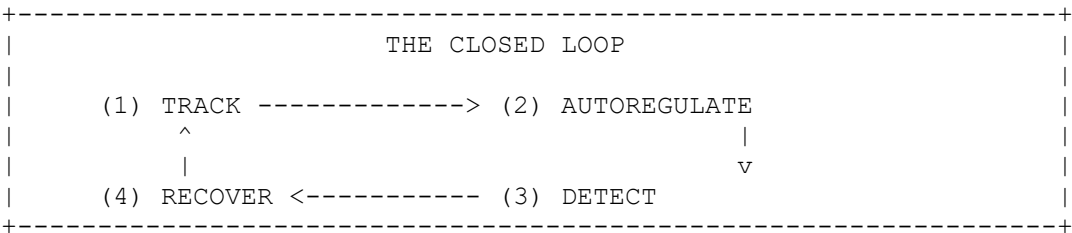
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1. Executive Summary & Core Value Proposition

FORGE (also referenced in research dossiers as *IRONLOOP*) is India’s first integrated, closed-loop Velocity-Based Training (VBT) hardware and software ecosystem. Designed to eliminate non-contact overuse injuries and optimize resistance training, the system replaces arbitrary percentage-based load prescriptions with real-time, objective physiological feedback.

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The Closed-Loop Workflow

1. **TRACK:** Real-time barbell/machine velocity and acceleration via a magnetic inertial measurement unit (IMU) streaming at high data rates via Bluetooth Low Energy (BLE).

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2. **AUTOREGULATE:** Dynamic daily load selection using statistical load-velocity regression profiles derived from warm-up sets, eliminating risky 1-Repetition Maximum (1RM) testing.

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3. **DETECT:** Immediate within-set fatigue flagging (via velocity-loss thresholds) combined with smartphone camera pose estimation to identify technique breakdown and movement asymmetries.

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4. **RECOVER:** Explainable, rule-based return-to-training guidance and movement-correction protocols linked to persistent performance or technique flags under coach or professional oversight.

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Key Market Differentiator

Commercial gold-standard VBT units (e.g., GymAware, Output Sports) cost between ₹1.5L and ₹4L (\$500–\$2,500+ USD). FORGE delivers a Made-in-India hardware-software solution targeted at **₹5,000–₹8,000**, bridging the price gap for over 20 million Indian gym members, 9.2+ lakh Khelo India athletes, and grassroots S&C coaches.

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2. Problem Statement Analysis & Research Foundation

2.1 Problem Analysis (SIH26213 Context)

Under AICTE/MIC Problem Statement **SIH26213** ("*Ideas that can boost fitness activities and assist in keeping fit*"), FORGE solves three core issues in resistance training:

- **Invisible Fatigue Accumulation:** Athletes frequently feel capable despite experiencing acute neuromuscular fatigue, leading to tissue stress and overuse injuries.
- **Flawed Load Prescriptions:** Traditional percentage-based training (e.g., "70% 1RM") fails to account for daily fluctuations in readiness caused by sleep, stress, or nutrition.
- **Unstructured Recovery:** When joint discomfort or form degradation occurs, lifters either stop training entirely or resume too early without objective return-to-training criteria.

2.2 Peer-Reviewed Scientific Foundations

The FORGE engineering architecture is anchored in peer-reviewed sports science literature:

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- **Injury Epidemiology in Strength Sports:**
 - *Aasa et al. (2017)*: Injury prevalence in powerlifting ranges from **1.0 to 4.4 injuries per 1,000 training hours**, predominantly affecting the lower back, knees, and shoulders.

- *Keogh & Winwood (2017)*: Across strength sports, injury rates span **1.0 to 5.4 per 1,000 hours**, with the vast majority classified as chronic overuse rather than acute traumatic events.
- *Updated Systematic Evidence (2025)*: Weightlifting competition period prevalence ranges from **10.7% to 68%**, with training incidence rates between **2.4 and 3.3 injuries / 1,000 hours**.
- **Training Load & Injury Relationship:**
 - *Drew & Finch (2016) & IOC Consensus Statement*: Rapid training load spikes and unmanaged load accumulation represent primary modifiable risk factors for injury and illness.
- **Velocity as Load Intensity Indicator:**
 - *González-Badillo & Sánchez-Medina (2010)*: Mean Concentric Velocity (MCV) strongly correlates ($r>0.95$) with relative loading intensity (%1RM) across individuals, enabling safe load prescription without maximal testing.
- **Neuromuscular Fatigue & Velocity Loss:**
 - *Sánchez-Medina & González-Badillo (2011)*: Velocity loss during a set serves as a direct real-time signal of neuromuscular fatigue. Sets stopped at **20% velocity loss** achieve strength adaptations comparable to failure sets while incurring a fraction of the metabolic stress and recovery lag.
- **e1RM Prediction via Load-Velocity Profiling:**
 - *Banyard et al. (2017)*: Linear regression using submaximal loads predicts 1RM back squat with high reliability ($ICC>0.97$) and accuracy within $\pm 3\text{--}5\%$ of actual 1RM.

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Minimum Velocity Thresholds (MVT) Utilized by FORGE

These literature-established lower velocity limits define maximal effort across major movements:

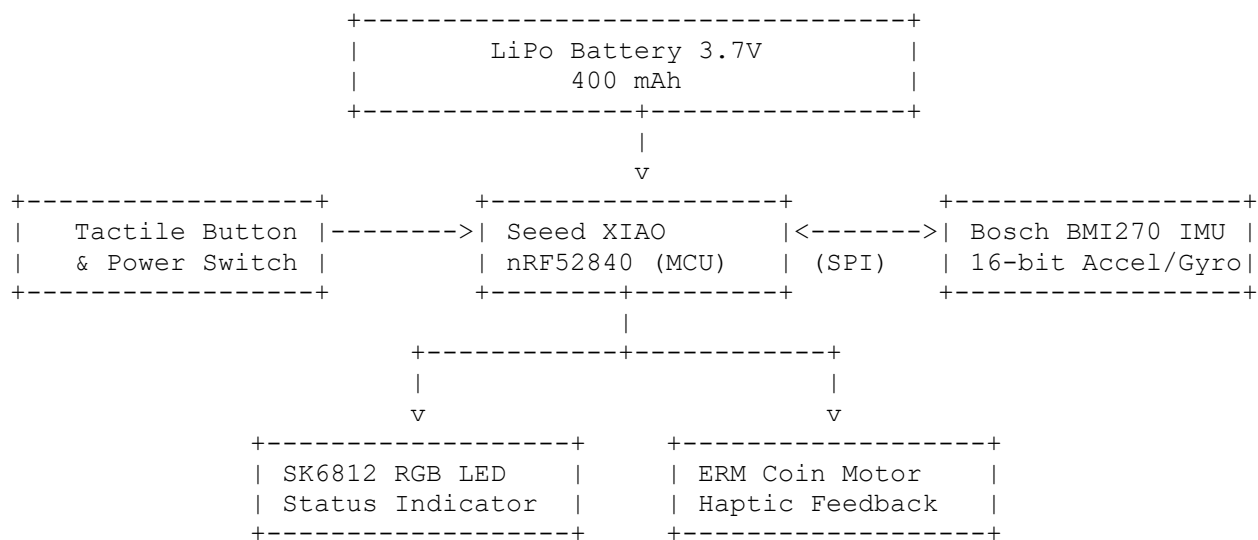
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Exercise	Minimum Velocity Threshold (MVT)	Literature Reference
Back Squat	0.32±0.05 m/s	García-Ramos et al. (2018)

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Exercise	Minimum Velocity Threshold (MVT)	Literature Reference
Bench Press	0.16±0.04 m/s	González-Badillo et al. (2017) PDF
Deadlift	0.23±0.06 m/s	Garnacho-Castaño et al. (2015) PDF
Overhead Press	0.22±0.05 m/s	Jiménez-Alonso et al. (2022) PDF

3. Hardware Architecture & Hardware Bill of Materials



3.1 Hardware Subsystems

- **Main Controller:** Sseed Studio XIAO nRF52840 (64 MHz ARM Cortex-M4F, Bluetooth 5.0 LE @ 2 Mbps, 1 MB Flash, 256 KB RAM, integrated LiPo charger).

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- **Inertial Sensor:** Bosch BMI270 6-axis IMU over SPI (10 MHz).

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- *Noise Density Comparison:* BMI270 accelerometer noise is **83 µg/Hz** versus 400 µg/Hz on MPU-6050. Lower sensor noise minimizes double-integration drift during kinematic tracking.

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- **Single MCU Justification:** A single nRF52840 easily handles SPI sampling at 200 Hz, Madgwick orientation updates, and BLE streaming, avoiding extra BOM cost, board real estate, and synchronization complexity.

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- **User Interface & Enclosure:**
 - **Tactile Switch:** 6mm button for manual set trigger and calibration.

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- **RGB LED & Haptics:** SK6812 mini RGB LED and an ERM coin vibration motor (1027) for real-time state indication and rep count confirmation.

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- **Magnetic Mount:** Integrated N52 Neodymium disc magnet (20 mm×3 mm, ~3 kg pull force) flush-mounted to a PETG 3D-printed shell (40×28×16 mm). Magnetic release prevents mechanical damage if the barbell is dropped.

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3.2 Complete Prototype Bill of Materials (BOM)

Component	Specifications / Model	Qty	Unit Cost (₹)	Total Cost (₹)	Source
Microcontroller	Seeed Studio XIAO nRF52840	2	1,200	2,400	Robu.in / Amazon
					PDF
IMU Sensor	Bosch BMI270 Breakout	2	300	600	Mouser / Local
					PDF
LiPo Battery	3.7V 400 mAh (402040)	2	220	440	Robu.in
					PDF
Neodymium Magnet	N52 Disc (20 mm×3 mm)	6	80	480	Amazon India
					PDF
Power & User Controls	Slide Switch + Tactile Buttons	6	-	50	Local Electronics
					PDF
Indicators & Haptics	SK6812 RGB LED + ERM Motor	6	-	170	Local / Robu.in
					PDF

Component	Specifications / Model	Qty	Unit Cost (₹)	Total Cost (₹)	Source
Passives & Discrete	Transistors, Diodes, Resistors, Caps	Misc -		95	Local Electronics PDF
Prototyping PCB	5x7 cm Perfboard + JST Connectors	6	-	200	Local Electronics PDF
Mounting Hardware	Steel Adapter Disc + Silicone Band	2	150	300	Local Fabricator PDF
Enclosure	3D Printed PETG Housing	2	300	600	Engineering Lab PDF
Wiring & Miscellaneous	28 AWG Silicone Wires, Solder	1	100	100	Local PDF
Development Tool	SEGGER J-Link EDU Mini (One-time)	1	2,800	2,800	Mouser India PDF
TOTAL PROTOTYPE COST	<i>(Includes 2 units + Dev Tools)</i>			₹8,235	<i>(Per-unit BOM: ~₹3,100)</i> PDF

4. Firmware & Kinematic Computation Pipeline

4.1 RTOS Framework & GATT Design

Firmware is implemented on **Zephyr RTOS** (C/C++), separating data acquisition, signal filtering, and BLE tasks into prioritized threads:

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- **Acquisition Thread (Priority 1 - Highest):** Reads BMI270 FIFO buffer at 200 Hz via SPI.

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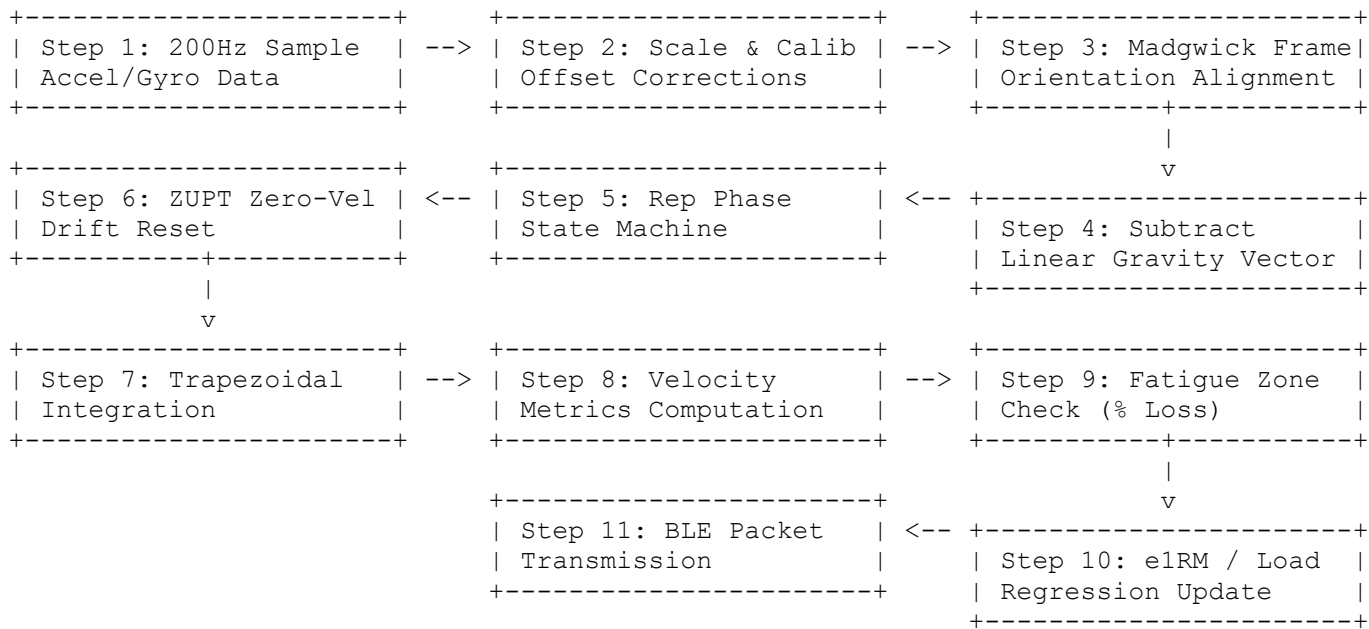
- **Processing Thread (Priority 2):** Executes Madgwick filter fusion, gravity removal, ZUPT checks, and rep integration.

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- **BLE Thread (Priority 3):** Handles GATT service notifications (Live Rep Data, Session Summary, Status, Commands).

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4.2 11-Step Kinematic Computation Pipeline



- 1. **Sampling:** 16-bit Accelerometer ($\pm 16g$) and Gyroscope ($\pm 2000^\circ/s$) raw acquisition at 200 Hz.

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- 2. **Calibration:** Apply static offset bias corrections.

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- 3. **Orientation Tracking:** Compute sensor orientation using Madgwick gradient descent algorithm (2° RMS error).

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- 4. **Gravity Isolation:** Transform accelerometer readings to the global frame and subtract 1g vertical acceleration.

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- 5. **Phase Detection:** Identify transitions between stationary, eccentric, and concentric phases.

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- 6. **Zero-Velocity Update (ZUPT):** Reset velocity drift to 0 m/s during stationary pauses between reps.

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- *ZUPT Effectiveness*: Reduces velocity drift per rep from $\sim 0.30\text{--}0.50$ m/s down to $\sim \mathbf{0.01\text{--}0.03}$ m/s.

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7. **Trapezoidal Integration**: Integrate vertical linear acceleration over the concentric duration:

$$v(t) = v(t-1) + T \cdot a(t-1) + \frac{1}{2} T^2 [a(t) - a(t-1)]$$

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8. **Metric Extraction**: Derive Mean Concentric Velocity (MCV), Peak Concentric Velocity (PCV), and Concentric Duration.

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9. **Fatigue Evaluation**: Calculate within-set velocity loss percentage relative to the set's fastest rep:

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$$\text{Velocity Loss \%} = \frac{v_{\text{fastest}} - v_{\text{current}}}{v_{\text{fastest}}} \times 100$$

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10. **Load Engine Update**: Map submaximal MCV against historical velocity profiles.

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11. **BLE Transmission**: Stream structured rep packets to the mobile app.

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5. Software Architecture & Intelligence Engines

5.1 Mobile Application Stack

- **Framework**: Flutter (Dart) with Riverpod state management.

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- **Local Database**: Isar DB (offline-first local persistence for fast, low-latency session capture).

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- **Backend Cloud Sync**: Firebase Firestore and Authentication (synchronizes weekly summaries and coach views).

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- **Visualization:** `fl_chart` for velocity profiles and `e1RM` trend graphs.

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MOBILE APP LAYER (FLUTTER)		
Live Session View (BLE Stream & Charts)	Camera Analysis (ML Kit Pose View)	Coach / Athlete Dashboard (Isar DB / Firestore)
Load Autoregulation Engine (Linear Regr.)	Camera Pose Module (Angle & Depth Math)	Rule-Based Rehab Engine (curated rehab_db.json)

5.2 Load Autoregulation & Velocity Zones

The app performs dynamic linear regression ($\text{Load} = a \cdot \text{MCV} + b$) on warm-up sets to prescribe daily working loads according to target RPE:

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Velocity Zone	Velocity Range (m/s)	Relative Load (%1RM)	Target RPE	Training Adaptation
Speed	>1.00	<50%	1–3	Neuromuscular Speed & CNS Prep
Power	0.75–1.00	50–70%	4–6	PDF Explosive Power & Dynamic Effort
Strength-Speed	0.50–0.75	70–85%	6–8	PDF Hypertrophy & Main Strength Sets
Maximal Strength	0.35–0.50	85–95%	8–9	PDF Heavy Strength Prescriptions
Near-Maximal	<0.35	>95%	9–10	PDF Peak Testing (Infrequent)
				PDF

Zones derived from González-Badillo & Sánchez-Medina (2010) back squat data.

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5.3 Computer Vision Form Analysis Module

Utilizing Google ML Kit Pose Detection, the camera pipeline estimates 33 body landmarks offline at 20–30 fps:

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- **Squat Depth Check:** Verifies hip landmark vertical displacement relative to knee joint level.

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- **Knee Valgus Flag:** Quantifies medial knee displacement against the hip-to-ankle axis (>15% femur length displacement triggers a valgus flag).

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- **Asymmetry Tracking:** Measures left-right joint angle variances during bilateral lifts.

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5.4 Curated Rule-Based Rehab Engine

FORGE avoids non-deterministic "black-box" generative AI for health claims, utilizing an explainable, expert-authored 3-layer architecture:

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1. **Knowledge Layer (`rehab_db.json`):** Version-controlled database of corrective exercises, execution cues, and progression criteria.

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2. **Trigger Logic:** Requires a movement or fatigue flag to persist across **N consecutive reps/sessions** before firing, suppressing single-rep false positives.

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3. **Presentation & Escalation:** Renders actionable correction cards alongside clear medical disclaimers and flags red-flag symptoms for clinical escalation.

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6. Competitive Positioning & Market Analysis

6.1 Feature & Price Matrix

Feature / Metric	GymAware RS	Output Sports	Vitruve / RepOne	GPath	FORGE (IRONEYES)
	PDF+ 1	PDF+ 1	PDF	PDF	PDF
Sensing Technology	Linear Transducer	IMU Wireless	Linear Encoder	IMU Clip	Barbell IMU + Vision
	PDF	PDF	PDF	PDF	PDF+ 1
Real-time VBT	Yes	Yes	Yes	Yes	Yes
	PDF	PDF	PDF	PDF	PDF
e1RM Tracking	Yes	Yes	Partial	Basic	Yes
	PDF	PDF	PDF	PDF	PDF
Fatigue Monitoring	Yes	Yes	No	Basic	Yes (Real-time)
	PDF	PDF	PDF	PDF	PDF
Camera Form Analysis	No	No	No	No	Yes (ML Kit)
	PDF	PDF	PDF	PDF	PDF+ 1
Integrated Recovery	No	No	No	No	Yes (Rule-based)
	PDF	PDF	PDF	PDF	PDF+ 1
India Support / Focus	Imported	Imported	Imported	Imported	Made in India
	PDF+ 1	PDF+ 1	PDF	PDF	PDF
Retail Unit Price	₹1.5L – ₹2.5L	₹2.0L – ₹4.0L/yr	₹25k – ₹40k	~₹11,500	₹5,000 – ₹8,000
	PDF	PDF	PDF	PDF	PDF

6.2 Serviceable Addressable Market (SAM) & Business Model

- **Target Audience:** Competitive athletes (~50,000), serious recreational lifters (~2–3M), S&C coaches (~5,000 facilities), and Khelo India / SAI sports hostels.

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- **Year 1–3 Revenue Estimate:** ₹10 – ₹20 Crore Serviceable Addressable Market.

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- **Revenue Streams:**

- **Hardware Sales:** Retail target ₹6,999 (~70% gross margin based on production BOM).

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- **Software SaaS Subscriptions:** Free tier for basic tracking; Athlete Pro (₹299/mo); Coach Plan (₹999/mo up to 20 athletes); Academy Plan (₹2,499/mo).

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7. Technical Feasibility, Risk Analysis & Defense Strategy

7.1 Risk Matrix

Risk Factor	Severity Likelihood		Technical Mitigation Strategy
IMU Double Integration Drift	High	High	Combine Madgwick orientation estimation with Zero-Velocity Updates (ZUPT) to bound velocity error to <0.03 m/s per rep. PDF+ 1
BLE Signal Droppage in Gyms	Medium	Medium	Store rep logs locally on hardware flash memory buffer; automatically resync missing sequence packets upon reconnection. PDF+ 1
Overclaiming Medical AI Capabilities	High	Medium	Explicitly frame algorithms as statistical regression, geometry calculations, and curated expert rules. Avoid generic "AI" claims. PDF+ 2
Liability from Rehab Protocols	High	Low	Enforce persistent disclaimers, require N-rep trigger persistence, and refer acute pain symptoms directly to medical care. PDF+ 2

7.2 Jury & Defense Attack Points

PREPARED JURY DEFENSES	
JUDGE: "IMUs suffer from drift. How can you match linear position transducers?"	
DEFENSE: FORGE uses ZUPT during inter-rep pauses to reset integration error, reducing per-rep drift to <0.03 m/s. For fatigue tracking, detecting relative velocity loss percentage is what matters, not absolute lab precision.	
JUDGE: "How can you claim this device prevents injuries?"	
DEFENSE: FORGE manages modifiable risk factors by flagging velocity loss and training load spikes. We provide training-load intelligence, not a guaranteed medical diagnosis.	

| JUDGE: "Where is the complex Machine Learning in your system?"|
| DEFENSE: Our core systems rely on peer-reviewed linear regressions and auditable|
| expert rules. This explainability guarantees that every recommendation is|
| transparent, safe, and verifiable by coaches.|
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8. Social, Economic, & Government Policy Impact

- **Social Impact:** Democratizes sports science by providing SAI-level performance monitoring tools to student-athletes across India's 800+ Khelo India Centres at an accessible price.
- **Economic Impact:** Replaces expensive imported equipment (saving ₹1.5L–₹4L per unit). Preventing non-contact injuries like ACL tears saves athletes between ₹2L and ₹5L in surgical and rehabilitation expenses.
- **Government Alignment:**
 - **Atmanirbhar Bharat:** 100% indigenous hardware and software development.
 - **Khelo India & Fit India Movement:** Directly supports athlete monitoring and injury prevention directives outlined in SAI's National Sports Science agenda.

9. System Implementation Roadmap

Phase 1: Sept 2026	Phase 2: Oct-Nov 2026	Phase 3: Nov-Dec 2026	Phase 4: 2027
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Hardware Prototype	Load Engine &	ML Kit Form Analysis	Mass Production PCB,
& BLE Velocity Stream -->	Fatigue Detection -->	& Curated Rehab Rules -->	Retail Launch &
[COMPLETED]	[IN DEVELOPMENT]	[DESIGNED]	Khelo India Integration
+-----+	+-----+	+-----+	+-----+

10. Recommended SIH Presentation Structure (12 Slides)

- SLIDE 1: Title & Hero Render
 - Title: FORGE - Train with Intelligence. Prevent Injury. Recover Faster.
 - Subtitle: SIH26213 | Team IRONEYES | Fitness & Sports (Hardware)
 - Visual: High-contrast product hero render with physical prototype on table.
- SLIDE 2: Problem - India's Gym Users Train Blind
 - Key Data: 3.3 injuries / 1,000 hours in strength sports; 38% overreaching rate.
 - Visual: Training timeline showing undetected load accumulation leading to injury.
- SLIDE 3: The 3 Root Causes
 - Points: Fixed % prescriptions ignore daily readiness; fatigue is invisible; no affordable rehab.
 - Visual: Three clean icon callouts highlighting readiness, fatigue, and recovery gaps.
- SLIDE 4: The Solution - FORGE Closed Loop
 - Solution: Track -> Autoregulate -> Detect -> Recover in one device for ₹5,000.
 - Visual: Interconnected 4-node closed-loop diagram.
- SLIDE 5: Live Hardware Demo
 - Action: Live demonstration streaming barbell reps, velocity loss %, and red fatigue flags.
 - Visual: "FORGE - LIVE DEMO" screen backdrop with real-time telemetry metrics.

SLIDE 6: Hardware Architecture
 |-- Specifications: Saeed XIAO nRF52840, Bosch BMI270, ZUPT drift correction, PETG casing.
 --- Visual: Annotated hardware cross-section diagram.

SLIDE 7: Dual Intelligence Engines
 |-- Features: On-device real-time fatigue zones; explainable rule-based rehab database.
 --- Visual: Two-column layout comparing load regression and recovery logic.

SLIDE 8: Camera Form Analysis
 |-- Features: ML Kit joint tracking (33 landmarks), depth validation, valgus detection.
 --- Visual: Skeleton overlay on squat execution with real-time angle metrics.

SLIDE 9: Competitive Matrix
 |-- Comparison: Highlights FORGE's price advantage (₹5,000 vs. GymAware's ₹1.5L+).
 --- Visual: Feature comparison table with FORGE highlighted across all criteria.

SLIDE 10: Social, Economic & Strategic Impact
 |-- Impact: Import replacement, ACL injury cost savings, alignment with Khelo India/Atmanirbhar.
 --- Visual: Impact flywheel with logos for Khelo India and Fit India.

SLIDE 11: System Development Roadmap
 |-- Timeline: Phase 1 (Built) -> Phase 2 (In Development) -> Phase 3/4 (Planned).
 --- Visual: Four-phase timeline with status badges for each milestone.

SLIDE 12: Team Credentials & Call-to-Action
 |-- Details: Team IRONEYES credentials, technical roles, coaching experience.
 --- Visual: Team contact details, QR code to demo video, and physical prototype presentation.

11. Complete Reference List

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